



Indoor Signage Best Practices

An Analytical Report Generated by the AI Research Council

Transform Airports — AI Research Council

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FINAL — FOR DISTRIBUTION

Abstract

Wayfinding is the cheapest operational lever in an airport and the one most consistently squandered by design committees. The airports that wayfind well (SIN, ZRH, MUC, HEL) treat signage as a system with explicit performance criteria; the airports that wayfind poorly (most US hubs) treat it as graphic design served family-style across multiple stakeholders. The gap is not budget — it's discipline about hierarchy, decision-point analysis, and refusing to let every stakeholder add their own sign. MWAA has accumulated decades of signage decisions at IAD and DCA that now contradict each other; a forward standard is overdue.

This report should treat 003 (sterile circulation) and 005 (baggage) as priors — wayfinding interacts with both. The report examines this claim across the following: The decision-point methodology: how high-performing airports actually map where passengers must make choices, and how they validate that signs answer the question being asked at that point; Sign hierarchy and information density: what the research (Passini, Arthur, Mollerup, more recent eye-tracking studies) actually says about glance time, character height, and message complexity. Distinguish established findings from designer folklore; Multilingual and pictographic strategy: where AIGA/ISO 7001 pictograms still hold, where they break, and how international hubs handle language layering without visual chaos; Dynamic vs. static signage: where digital displays earn their cost (gate changes, FIDS, baggage claim) and where they actively degrade wayfinding (cluttered concourse navigation); Governance: who at a well-run airport owns signage standards, what veto authority they hold against tenants and TSA, and how that authority is structured. This is the real lever.

How This Report Was Produced

This report is the product of the Transform Airports AI Research Council — a multi-agent analytical system running on Anthropic's Claude models. Rather than a single model writing in one pass, the Council convenes a panel of specialized research agents, subjects their work to adversarial review, and verifies every claim before publication. The process runs in four stages, with human review at two checkpoints.

Stage 1 — Independent research. Each research agent below produced an evidence brief without seeing the others' work, so the analysis draws on genuinely independent lines of evidence rather than a single line of reasoning.

Stage 2 — Synthesis and adversarial review. A Strategist synthesized the briefs into an argument; a Red Team attacked it across two rounds; the Strategist revised after each.

Stage 3 — Edit and fact-check. An Editor tightened the prose. A Fact-checker verified every number against the research briefs, removing or flagging what it could not source.

Stage 4 — Production. The verified draft was formatted into this document.

Research agents seated for this report

- Airline Commercial Strategist
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Process agents (on every report)

- **Strategist** — synthesized the research briefs into a single argumentative draft.
- **Red Team** — attacked each draft to surface weak claims and logical gaps.
- **Editor** — tightened the prose and removed filler without adding new content.
- **Fact-checker** — verified every numerical and attributed claim against the briefs.

Each agent is governed by a versioned specification — its analytical mandate, evidentiary standards, source discipline, and argumentative posture. The complete technical definitions for every agent on the Council are published and openly available at:

<https://github.com/TransformAirports/ai-council/tree/main/.claude/agents>

The Cheapest Lever

A Report on Airport Wayfinding as a Governance Discipline

Transform Airports AI Council — Indoor Signage Best Practices Run Stage 3 — Final Draft (Editor + Fact-Check) June 3, 2026

Handoff notes

Weaknesses retained in this draft, and why:

- **The Section VI cost range is an order-of-magnitude estimate.** The briefs do not contain clean Tampa, Denver, or PANYNJ implementation cost numbers at program scale. The honest move is to label the figure a range and tell the reader what it rests on.
- **The eye-tracking evidence base is thin.** Heathrow n=108, the 2025 lab study n=60, Munich four days of data. The argument leans on convergence across the three studies; if a reader rejects the evidence base on small-n grounds, the case for wayfinding-as-governance still holds on institutional and historical grounds alone.
- **The \$700M concession-loss figure is cited once, in Section V, framed as a number to set aside.** Cutting it entirely would have been cleaner; it is retained once because readers will have heard it from vendor decks, and explicit dismissal is more useful than silent absence.
- **The opening IAD scene is illustrative.** The briefs do not give a misconnect probability for D-gate connections at IAD [UNVERIFIED — HUMAN REVIEW]. The scene is framed with "Imagine"; the eye-tracking and decision-point research carry the empirical load.
- **Frankfurt 1972 is supporting evidence, with the new-terminal qualifier stated.** PANYNJ is the present-day retrofit case the argument actually depends on.

I. The Decision Point

Imagine six in the morning at Washington Dulles. A passenger steps off a United mainline aircraft at Concourse C with a connection to Concourse D. She has thirty-five minutes — United's published minimum, the same number Delta promises at Atlanta. She has never been to IAD before. She walks up the jet bridge into the holdroom, past gate signs branded in United blue, and looks for the sign that will tell her what to do next.

The sign she needs has to answer a specific question. Not "where are the gates" — she knows where the gates are. Not "where is ground transportation" — she is not going to ground transportation. The question is: *which mode of underground transit will get me to Concourse D?* The answer is not the AeroTrain. The AeroTrain runs to Concourses A, B, and C. Concourse D is still served by Mobile Lounges — the two-story buses Eero

Saarinén designed in 1962, departing every five minutes, taking up to fifteen minutes to reach the concourse. United's own airport guide tells passengers this. The airport's signage, layered across decades of construction, tells them less reliably.

If she reads the right sign, she has twenty minutes of buffer. If she boards an AeroTrain, rides to Concourse C station, comes back, walks to the Mobile Lounge bay, waits — she misses the flight. The thirty-five-minute MCT is constructed for the best-case path [Airline Commercial Strategist Brief]. The signage at the decision point determines which path she takes. The signs at IAD were installed under at least three different design standards across at least three construction eras. Some are 1990s vintage. Some came in with the AeroTrain in 2010. Some arrived with the Silver Line in 2022. They do not agree with each other about colors, hierarchies, terminology, or what the passenger is presumed to already know.

One passenger, one decision point, one US hub, one morning. The institutional question is what happens when no one owns the standard that resolves it.

Wayfinding is the lowest-cost, highest-impact operational intervention available to a US hub airport. The reason most US hubs wayfind poorly is not budget. It is the absence of a governance structure with authority to enforce a single hierarchy against airlines, federal agencies, concessionaires, and the airport's own capital programs — each of which tends to add signs faster than the system can absorb them. The exemplars cited in every wayfinding case study — Changi, Munich, Zurich, Helsinki — wayfind well not primarily because they spent more, but because a single design authority owned the standard across decades of expansion, with explicit performance criteria, and refused to negotiate. MWAA, having accumulated three decades of contradictory signage decisions at IAD and DCA, has the opportunity to do the same. The standard exists. The question is whether the authority to enforce it does.

II. Executive Summary

Start with the asymmetry. MWAA's 2025–2039 Capital Construction Program totals \$9 billion — roughly \$600 million per year [Infrastructure Economist Brief]. A wayfinding program scaled to MWAA's traffic, derived from Heathrow's 2013 framework and inflated to 2026 dollars, sits at \$8 to \$13 million for a one-time effort — one to two percent of a single year's capital spend [Infrastructure Economist Brief; analyst calculation; derivation in Section VI]. That is the cost of governing how legible every square foot the rest of the program builds will be to the passenger walking through it.

The methodology was solved forty years ago and federally codified thirteen years ago. Romedi Passini's 1984 framework reframed wayfinding as spatial problem-solving: identify each point where the passenger must choose, provide the minimum information needed to resolve the choice. ACRP Report 52 (2011), adopted into FAA Advisory Circular 150/5360-12F (2014), codified three principles — continuity, connectivity, consistency — and gave continuity a testable number: no passenger should lose sight of a

directional cue across more than thirty meters of journey path [Chief Engineer Brief; Operations Analyst Brief]. The Advisory Circular is guidance, not regulation; compliance is not required for AIP grant eligibility [Regulatory and Political Analyst Brief]. Most US hubs have never been audited against it, because most cannot be — they do not maintain a documented decision-point map from which spacing could be measured.

The dynamic-versus-static debate is the wrong frame. Dynamic displays earn their cost where information is time-variant — flight status, gate changes, baggage carousel assignments, security wait times. Static signage earns its cost where information does not change — directional confirmation under stress at decision points. The Munich Airport eye-tracking study found that passengers largely ignored signage irrelevant to their journey. Information density that does not serve the current decision is not neutral noise: passengers learn to filter against it, and the filter generalizes [Operations Analyst Brief; Architectural Historian Brief]. App substitution does not close the gap. Only 7 percent of departing passengers use the airport-branded app [Technology Scout Brief]. The passenger who most needs help — the unfamiliar traveler, the elderly passenger, the connecting passenger under time pressure — looks up at the sign, not down at the phone.

The binding constraint is governance, not design vocabulary. At a typical US large hub, the airport authority controls common areas; airlines control leased gate areas; TSA controls the checkpoint zone; CBP controls the FIS hall; concessionaires control advertising positions. No federal regulation requires any of them to coordinate [Regulatory and Political Analyst Brief]. The cost of that arrangement is what the briefs collectively call contradiction debt. Pittsburgh is the cleanest case: built in 1992 as a US Airways connection hub, signed for the connecting passenger; the hub collapsed in 2004; the wayfinding system stayed connection-optimized for twenty years until a \$1.5 billion greenfield terminal opened in November 2025 [Aviation Historian Brief]. MWAA's terminals are in Chapter 2 of the same story.

Two execution constraints sharpen the implications. Airport construction runs forty to fifty percent above conventional commercial work because it schedules around flight operations; Ontario International budgeted a signage project at \$485,000 and finished at \$1.3 million, a 168 percent overrun [Chief Engineer Brief; Infrastructure Economist Brief]. The IAD Main Terminal carries National Register of Historic Places designation; overhead pendant signs in the Saarinen catenary-cable roof require structural analysis and preservation review on every move [Chief Engineer Brief]. MWAA's current allocations — \$514,000 for the IAD Signage and Wayfinding Masterplan, \$4.87 million for the DCA equivalent — are study budgets, not implementation budgets [Chief Engineer Brief]. The 2025–2039 Airport Use and Lease Agreement, a hybrid residual structure, gives United (approximately two-thirds of IAD enplanements including United Express) and American (53 percent of DCA flights) practical influence over any capital improvement that enters the rate base [Airline Commercial Strategist Brief]. A wayfinding program structured to keep capital-classified spend below the majority-in-interest threshold — with the operating-standard component carrying most of the cost — can be advanced without triggering formal MII review. The classification itself will be a negotiation.

III. The Argument

The Decision Point Is the Unit of Analysis

The fundamental act of bad airport signage is to answer a question the passenger is not asking. Romedi Passini, the Canadian environmental psychologist who wrote *Wayfinding in Architecture* in 1984, framed it precisely. A passenger entering a terminal is not executing a pre-loaded route. She is solving a series of nested problems. Each requires information at a specific moment and a specific place. The job of the sign system is not to publish the airport's full directory; it is to answer the next question.

Passini reframed wayfinding from aesthetics to operations: what does the passenger need to decide *at this point*, and does this sign answer that? The Danish designer Per Mollerup, who built the wayfinding systems for Copenhagen, Oslo, and Stockholm Arlanda between 1989 and 1998, coined "wayshowing" to drive the distinction home: wayfinding is what the passenger does; wayshowing is what the environment does. Design committees, Mollerup observed, optimize for the wrong variable — beauty or brand rather than the specific navigation decision the sign must resolve [Architectural Historian Brief].

Mollerup's ninth wayfinding strategy — social navigation, asking a staff member — is the failure mode. Every time a passenger approaches a customer service agent for directions, the environment has failed to answer a question it should have answered without speech. Industry data — vendor-sourced and not independently audited — places the share of customer service agent time consumed by directional questions at roughly eighty percent at large hubs; the number is directional, not audited, and should be treated as a proxy for system failure [Operations Analyst Brief]. The harder evidence is the ACI ASQ data: across the global Airport Service Quality survey, wayfinding ranks consistently in the top six satisfaction drivers, and at the top eight US gateway airports surveyed for ACRP Report 161, finding their way was the leading concern passengers reported above amenity satisfaction [Architectural Historian Brief; Operations Analyst Brief]. Wayfinding is not a background variable. It is what the passenger remembers.

In 2011, the Transportation Research Board published ACRP Report 52 — *Wayfinding and Signing Guidelines for Airport Terminals and Landside*. The report operationalized Passini's framework. Identify each segment of the passenger journey. Map the decision points. At each, define the minimum information required to resolve the choice. Enforce three rules: continuity (no passenger loses sight of a directional cue), connectivity (cues chain across the full journey), consistency (the same system, the same hierarchy, throughout). In 2014, the FAA adopted ACRP 52's substance into Advisory Circular 150/5360-12F — *Airport Signing and Graphics*. (AC 150/5340-18H covers airfield marking only — runways and taxiways — and is a common confusion in scope documents; the two ACs address entirely different environments [Chief Engineer Brief].)

Continuity is testable at a specific number: no passenger should lose sight of a directional cue across more than thirty meters of their journey path. Sign spacing of 47 meters satisfies roughly 80 percent of passenger information demand; at 56 meters, demand satisfaction drops to 60 percent [Aviation Historian Brief; Chief Engineer Brief]. These are auditable numbers. Most US hub airports have not been audited against them, and most cannot be — they do not maintain a documented decision-point map from which spacing could be measured. Tampa International's Wayfinding Standards Manual, a 482-page document developed with Gresham Smith, includes a Journey Segment Evaluation Tool requiring every new capital project to be evaluated against the existing sign hierarchy at each decision point it touches, so individual project decisions cannot silently degrade the system [Operations Analyst Brief]. Tampa is operating ACRP 52 as a checklist. Most US hub airports are not.

What Eye-Tracking Has Actually Told Us

In December 2022 and January 2023, Heathrow Terminal 5 ran a formal eye-tracking study using Tobii Pro Glasses 3 on 108 participants moving through arriving, departing, and connecting journeys. This is the best public-domain evidence available on airport signage behavior. The participant count is small; the studies are commercial partnerships with Tobii. With those caveats stated once: what the study found is embarrassing to designers and operators alike.

The failures concentrated at transitional points — escalator exits, entries to large halls, multi-path junctions. Not the mid-corridor sections where most airports place their largest signs. Two failure modes dominated. First: signs that passengers simply did not notice, often because they were positioned outside the natural sightline at the moment the decision had to be made. Second: oversaturation at the same point, where competing information increased cognitive load and reduced the probability that the passenger registered any of it [Technology Scout Brief; Operations Analyst Brief]. Heathrow acted on the data, decluttered the decision points, reconfigured the immigration entrance, and revamped onward travel signage. The terminal's wayfinding satisfaction score, which had declined during COVID when operational necessity prompted sign proliferation, reversed after the intervention.

The Munich Airport study, using the same methodology over four days with real travelers, reached a finding that ought to be hung on the wall of every airport design office: *passengers largely ignored information on navigation signage that wasn't relevant to their journey*. Information density that does not serve the current decision is not neutral noise; passengers learn to filter against it, and the filter generalizes. A passenger trained to filter irrelevant content on one sign is more likely to filter relevant content on the next [Architectural Historian Brief; Operations Analyst Brief]. The system that adds more signs to communicate more information communicates, on average, less.

A 2025 controlled laboratory study in the *Journal of Asian Architecture and Building Engineering* — 60 participants varying information density, format, and time pressure — confirmed the mechanism. High information density and text-only formats increased cognitive load and reduced decision confidence; text-plus-graphic formats delayed

cognitive overload. Passengers under time pressure showed physiologically measurable strain — larger pupil diameters, lower decision confidence [Operations Analyst Brief]. The same sign that performs adequately for a leisurely traveler becomes actively misleading for a connecting passenger with twenty minutes.

Manchester Airport's documented improvement, under the direction of a single Head of Visual Environment with a clear directive to reduce sign volume, can be summarized in her own five words: *we're vastly reducing the number of signs on site* [Technology Scout Brief]. The lesson is not subtle. Most US hubs have learned the opposite one.

The Wayshowing Distinction and the Committee Problem

Mollerup's reframing locates the failure at the design committee. A signage committee, by disposition, optimizes for stakeholder representation. The airline operations team wants its boarding information visible. The retail tenant wants directional signage to its space. TSA wants its prohibited-items notice prominent. CBP wants its currency declaration warning. The fire marshal wants the egress signs lit. The ADA coordinator wants tactile and Braille requirements met at every doorway. The aviation marketing office wants the airport name and city brand visible. Each stakeholder is acting reasonably. None, by definition, is responsible for the cumulative cognitive load on the passenger.

This dynamic explains why every US hub airport's signage looks roughly the same — additively dense, hierarchically flat, visually crowded — even though every US hub has hired some of the same wayfinding consultants. The consultants produce competent design. The committees absorb it and then add to it. The system degrades to its governance floor: the weakest enforcement point becomes the system's effective standard.

The international exemplars — Frankfurt (1972), Singapore Changi (1981), Munich (1992), Zurich, Helsinki — share a feature the US largely lacks: a single airport authority with explicit veto power over commercial, federal-agency, airline tenant, and concessionaire signage. Otl Aicher's 1972 Frankfurt system embedded the rule in the airport's operating architecture: commercial advertising was spatially separated from wayfinding signage, on different surfaces, in different formats. The separation was enforced as a contract term, not negotiated case by case [Aviation Historian Brief]. The qualifier: Aicher built this discipline into a new terminal at the design stage, not as a retrofit onto an existing fragmented system. The case proves the contractual mechanism works when written into the building's program. It does not prove the same mechanism can be imposed on three decades of accumulated stratigraphy without political capital that no US authority has yet been willing to spend.

The Port Authority of New York and New Jersey attempted the closest US equivalent as a retrofit, commissioning Paul Mijksenaar to extend the Schiphol system across JFK, LGA, and EWR. The PANYNJ Airport Standards Manual for Pedestrian Signing and Wayfinding (2020) designates Aviation Customer Experience in the Central Office as the enforcement authority, with a Program Manager for Wayfinding and Connections

Solutions serving as principal reviewer of all signage submittals. The lease agreements bind terminal operators to compliance [Regulatory and Political Analyst Brief; Operations Analyst Brief]. The program required extraordinary political capital and remains in prototype rollout at JFK five years after the manual was finalized. PANYNJ is at once the proof that a US retrofit can be built and the proof that the work is slower and harder than any prior airport authority has appreciated. It is the case the argument actually depends on.

Contradiction Debt Is the Hidden Operating Cost

If the design problem was solved in 1972 and the methodology was federally codified in 2014, what does an unaddressed US wayfinding system look like in the field? A stratigraphy of layered decisions, each correct at the moment it was made, each contradicting the others now.

At Charlotte Douglas International, before remediation, passengers consistently overestimated walking times because inconsistent wayfinding cues induced uncertainty, which produced hesitation. The intervention — consolidating five alpha-coded security zones into three numeric zones, with color-coded portals and vertical beacons — reduced perceived complexity without physical reconstruction [Operations Analyst Brief]. The operational lesson runs counter to design instinct: *consistency is a separate variable from quality*. A uniformly mediocre system outperforms a partially excellent one where the excellent and mediocre elements contradict. A passenger who has learned the rules in one zone is being punished for that learning when she enters the next and the rules have silently changed.

Pittsburgh International is the cleanest US case of what contradiction debt costs across an operational regime change. The 1992 hub terminal was built for US Airways as a connection machine — an X-shaped airside concourse with 75 gates, every intersection equidistant from connecting flights, signage optimized for the connecting passenger under time pressure. The system worked for its stated purpose. In November 2004, US Airways collapsed the Pittsburgh hub. Enplanements fell from approximately twenty million to under eight million. The traffic profile inverted from sixty percent connecting to ninety-eight percent O&D [Aviation Historian Brief]. The wayfinding system remained connection-optimized. For two decades, PIT operated with signs calibrated to a hub that wasn't there. The cost is not directly measured in the public literature. The shape of it — a passenger navigating a near-empty connection concourse to find a single O&D departure gate — is exactly what the eye-tracking research predicts: hesitation, reconfirmation behavior, over-reliance on staff redirection. The system was not corrected until the \$1.5 billion greenfield terminal opened in November 2025. Two decades of mismatch. No mechanism for resetting it short of a billion-dollar capital event.

IAD carries six distinct construction events: the Saarinen Main Terminal (1962), the Midfield Concourses (1996), the underground walkway (2004), the AeroTrain (2010), the Concourse C expansion (2016), and the Silver Line Metro connection (2022) [Aviation Historian Brief]. Each event introduced new passenger circulation modes with no predecessor signage; each required new signage to explain a new option, layered onto

signage explaining earlier options that many passengers still used. By 2022, IAD passengers could choose among four modes to reach the concourses — mobile lounges, the underground walkway, the AeroTrain, and the Metro connection — each mode partially served by signage from a different construction era, in different fonts, with different terminology for the same destinations.

DCA is more compressed. Three terminal eras — Terminal A (pre-1941 and 1958), the César Pelli Terminal B/C (1997), and the post-9/11 emergency checkpoint zone — accumulated signage vocabularies produced by different designers under different standards [Aviation Historian Brief]. Project Journey, MWAA's \$1 billion DCA capital program, included a five-day window in June 2022 during which more than 1,000 signs were converted from terminal letter-names to terminal numbers and numeric gate designations to alphanumeric [Chief Engineer Brief]. The coordination required airline FIDS systems, boarding-pass systems, WMATA Metro station entrance signage, parking garage digital systems, and roadway signs to update on a synchronized timeline. An extraordinary operational achievement, and a reminder of how high the friction of any system-wide signage change has become.

What does the contradiction debt cost? Most airports cannot answer directly because no one routes operational exceptions back to the wayfinding system. Missed flights are logged against carrier delays. Gate holds for late passengers are logged against boarding inefficiency. Information desk volume is not tracked against wayfinding performance at most US airports. The measurement gap is itself a governance failure: if no one owns wayfinding as a performance system, no one measures its failures [Operations Analyst Brief].

What can be documented: a peer-reviewed 2024 study of 89 US airports found that a 10 percent increase in passenger dwell time was associated with a 6 percent increase in retail revenue and an 8 percent increase in food and beverage revenue [Infrastructure Economist Brief]. Dwell time has multiple drivers — security throughput, lounge access, retail mix, scheduling, wayfinding. Properly discounted, wayfinding-driven dwell-time improvement may capture twenty to thirty percent of the elasticity, which still leaves it as one of the few interventions that converts fixed terminal area into incremental non-aeronautical yield without renegotiating a lease or expanding a footprint.

For MWAA specifically, the yield gap is measurable. The airports reported combined in-terminal concession revenue of \$110 million in 2023 across approximately 52 million passengers — about \$2.12 per passenger, materially below the North American non-aeronautical average of \$8.68 per passenger (which includes parking and is not a perfect comparison) [Infrastructure Economist Brief]. At the brief's calibrated elasticity, a one percent improvement in dwell time implies \$660,000 to \$880,000 in incremental annual revenue. Allocating twenty percent to wayfinding's share implies \$130,000 to \$175,000 per percentage point. Multiplied by a plausible three- to five-point dwell-time gain across a multi-year program, the payback math is \$400,000 to \$875,000 annually. Modest, real, one component of a multi-stream return that also includes reduced

customer-service load, deferred sign maintenance, and reduced service-recovery exposure.

Why Digital Is Not the Answer the Vendors Sell

The dynamic-versus-static debate, as airports usually have it, is the wrong frame. The better frame, drawn from the eye-tracking literature, is to match each medium to the information-change rate of the location it serves.

Dynamic displays earn their cost where information is time-variant: flight information, gate changes, baggage carousel assignments, security wait times, concourse directories at hubs with frequent changes [Operations Analyst Brief; Technology Scout Brief]. The global FIDS market was \$527 million in 2024, growing at 7 percent CAGR — settled technology, no meaningful controversy [Technology Scout Brief]. Changi Terminal 2 retired its iconic split-flap boards in February 2024 after twenty-one years; nobody mourned the operational loss.

Static signage earns its cost where information does not change. Which direction is Terminal C. Where are the elevators. Decision-point under time pressure is exactly the moment when a static answer outperforms a rotating one. Denver International's 2013 overhaul understood this: it replaced 23 roadside and 22 overhead signs with retroreflective static alternatives, saving an estimated \$3 to \$4 million in avoided structural replacement costs while establishing pilot standards for digital displays in the terminal proper [Operations Analyst Brief; Chief Engineer Brief]. Retroreflective overhead signs are lower maintenance, lower power, and — critically — always on when there is a power event.

The lifecycle math is more sobering than capital programs typically recognize, and the implication is not "less digital." It is "fund the refresh discipline." Static backlit signs run a 15- to 20-year useful life with LED lamp replacement at years 8 to 10. Digital display panels in 24/7 airport environments run 5 to 7 years; media players run 3 to 5 years [Chief Engineer Brief]. A complete FIDS hardware-plus-software deployment runs \$8,000 to \$15,000 per display node, with premium gate-area units reaching \$20,000 to \$30,000. An estimated 35 to 40 percent of FIDS units installed at US commercial airports are operating beyond their stated lifecycle [Chief Engineer Brief]. Airports are carrying digital signage fleets installed in the early 2010s that have never been refreshed. The strategic error is funding digital installations in capital programs while treating refresh-cycle hardware replacement as an O&M problem, then deferring O&M when budgets tighten. The correct response is not less digital. It is digital funded with a published lifecycle plan and a refresh budget locked to the panel and media-player replacement cycle, governed by the same standards authority that owns the static system. Digital infrastructure without that discipline becomes a more expensive version of the contradiction debt it was supposed to solve.

The case for app-based wayfinding is more complex than the vendor pitch admits. The question is not whether the unfamiliar passenger has installed an indoor mapping product — Apple Maps is installed by default on every iPhone. The question is whether

the passenger under time pressure at a decision point will pull out the phone, open the app, identify her location, and read the directional overlay, or whether she will look up at the sign overhead. The Heathrow and Munich data record where passengers look under stress, and the answer is signs. The phone is for planning the journey before it starts and for tracking the bag after it ends. The escalator at the bottom of which she is making her decision is not phone work [Operations Analyst Brief; Technology Scout Brief].

The Governance Architecture That Actually Exists

The reason this debate keeps recurring at US airports is not that the designers have run out of ideas. It is that the governance arrangement makes the design irrelevant. The FAA has no enforceable authority over indoor terminal wayfinding; Advisory Circular 150/5360-12F is guidance, not regulation [Regulatory and Political Analyst Brief]. ADA creates a genuine regulatory floor, often implemented as a parallel sign layer that competes with the wayfinding hierarchy rather than integrating with it. TSA's Checkpoint Requirements and Planning Guide (May 2025) requires all signage within the checkpoint zone to be approved by TSA headquarters' Office of Signage Guidelines. CBP's Airport Technical Design Standards govern signage in Federal Inspection Services halls. Local fire marshals govern egress signage under NFPA 101. Concessionaires hold advertising display rights under contracts that typically run 10 to 15 years.

Three federal regulatory authorities govern the same square foot of terminal floor and do not coordinate with each other. The airport's own standard loses jurisdiction precisely at the two highest-stress moments in the passenger journey — security and customs [Chief Engineer Brief]. The contractual cage is the binding constraint: airline use and lease agreements grant carriers the right to brand their gate areas, ticket counters, and jetways, with airport approval authority that is broad in theory and rarely exercised in practice [Airline Commercial Strategist Brief; Regulatory and Political Analyst Brief].

The committee problem is the predictable output of distributed veto with no point of authoritative override. Naming it as such matters, because it locates the lever where it actually sits: not in design talent but in the institutional decision to centralize wayfinding authority and then defend that authority against every plausible objection from every stakeholder who will produce one.

IV. The Counter-Case, Honestly Presented

A good thesis does not benefit from straw-man opposition. The strongest case against the cheapest-lever argument deserves to be stated at full strength. The case combines three claims that mutually reinforce. A reader who accepts them is entitled to conclude that the cheapest-lever framing is, at best, an essay in the wrong decade.

The first claim is that the smartphone has changed the navigation environment in ways fixed signage cannot recover. The 7 percent app-usage number sounds like a navigation vacuum and is not. Apple Maps provides indoor mapping at more than thirty US airports, including Dulles, Reagan National, JFK, LaGuardia, Newark, LAX, SFO, SEA-

TAC, Denver, and Miami. Google's Live View indoor navigation, using the phone camera to overlay directional arrows in real time, is live at Zurich — one of the four exemplars on which the thesis depends — and at Sydney. IATA's 2025 Global Passenger Survey found that 78 percent of passengers want a single smartphone device to combine digital wallet, digital ID, loyalty cards, and airport navigation. Airline app usage has risen to 23 percent of passengers, up 7 percentage points in five years. The trajectory of passenger navigation is unambiguously digital and phone-centric. A wayfinding strategy in 2026 built around static sign hierarchy is optimizing for a passenger behavior model that is shrinking as a share of traffic [Contrarian Brief]. Sharper: Zurich's celebrated wayfinding reputation may owe part of its present-day clarity to the Google overlay rather than to the sign hierarchy alone. If true, the historical argument for governance-driven exemplars weakens accordingly.

The second claim is that the international exemplars are capital success stories the thesis is misreading as governance success stories. Helsinki's single-terminal clarity followed a €1 billion terminal expansion that consolidated all operations under one roof. Zurich's "Airport 2000" fifth construction phase cost CHF 2.4 billion and produced a purpose-built Airside Centre with unified hub geometry. Munich's Terminal 2 is a joint venture between Flughafen München GmbH and Deutsche Lufthansa AG. Changi is investing S\$3 billion over six years and S\$5+ billion in Terminal 5. The causal arrow runs the other way: these airports wayfind well because billions of dollars created unified physical environments in which a single design authority could operate without contractual contradictions. US airports cannot replicate the result without replicating the capital base. Spatial legibility cannot be retrofitted with signs. A spatially incoherent terminal will defeat any sign system [Contrarian Brief; Architectural Historian Brief].

The third claim is structural, and most binding. The single-authority governance model the thesis implicitly recommends does not exist under US lease structures. Thirty-six of the 66 largest US airports have majority-in-interest provisions that give signatory carriers the right to review and, at affirmative MII airports, to block capital projects above defined thresholds [Airline Commercial Strategist Brief]. Signage in airline-leased concourses is, by contract, airline signage. TSA controls checkpoint signage by federal mandate. CBP controls FIS signage by federal mandate. The FAA's own Advisory Circular 150/5360-12F draws an explicit distinction between "systems typically installed by an airport operator" and "those typically installed by individual airlines." A "single wayfinding authority with veto power over tenants, TSA, and airlines" is not a governance innovation US airports have failed to adopt out of managerial weakness. It is a legal construct that does not exist and cannot be conjured without renegotiating leases, federal agency memoranda, and security protocols simultaneously. The Port Authority's 2020 manual remains in prototype rollout at JFK five years later — not because PANYNJ lacks discipline, but because implementation runs through fourteen terminals, multiple federal partners, and dozens of carrier leases [Contrarian Brief; Regulatory and Political Analyst Brief].

Two corollaries sharpen the case. Capital renewal of aging signage is unavoidable, with or without governance reform. Tactile and Braille ADA components have a finite life;

backlit cabinets fail; pendant blades dent. The question is not whether capital will be spent on signage but whether it will be spent as patchwork inside individual capital projects or as a coherent program against a published standard. The alternative the thesis pretends does not exist is reactive spending, terminal by terminal, which is what every US airport already does. And staff-based interventions can deliver passenger experience improvements faster and cheaper than a sign rationalization, without crossing the MII threshold — PA tuning, gate agent positioning, information desk placement, and FIDS refresh-rate adjustments cost tens of thousands of dollars and can be deployed in weeks. A comprehensive signage rationalization at a major US hub involves cataloging thousands of existing signs (the Port Authority documented over 10,000 across its three-airport system), eight to twelve months of planning, fabrication of custom sign types in airport-rated materials, installation during active operations, and a permanent enforcement mechanism [Contrarian Brief].

The procedural punch lands last. If the Council publishes a finding that wayfinding dysfunction is a governance problem rather than a resource problem, the institutional takeaway at most US airports will be: form a committee. The airport director will convene a wayfinding working group that will produce a design standards document that will sit in a drawer while the airlines continue to hang their own signs and TSA continues to staple up its own stanchion-mounted notices. The governance-first framing becomes, in practice, an excuse for under-investment in systems that would actually work. It gives boards a low-cost way to appear to act — approve the style guide — while avoiding the appropriation that would produce results [Contrarian Brief].

A reader who follows this to its conclusion is entitled to two views. Either the thesis recommends an approach that depends on a governance structure that does not exist and cannot be built in a US contractual environment, or it recommends an approach boards will adopt as a fig leaf for inaction. Both versions take the cheapest-lever framing seriously. Both reject it.

V. Why the Counter-Case Is Insufficient

The counter-case is partly right. It is not right enough to overturn the thesis.

The smartphone trajectory is real. Apple Maps and Google Maps have done meaningful work to extend indoor navigation into airports; the IATA preference data is what it is. A wayfinding strategy that ignores the trend is foolish. But the substitution claim has two failures the thesis turns on. The population for which substitution holds — frequent flyers comfortable with phone-based navigation, often with the airline app installed, familiar with the specific airport — is the population that does not generate the wayfinding cost. The cost is generated by the unfamiliar traveler, the first-time international visitor, the elderly passenger, the connecting passenger six time zones from home. The phones are installed; the maps are present. The behavior at the decision point is not. The eye-tracking studies record where passengers look under stress, which is at signs. The mechanism the studies document — that stressed passengers reduce cognitive

load by fixating on the most immediate orientation cue — predicts sign preference at the exact moment the unfamiliar passenger would otherwise need to navigate by phone. The substitution argument is, at core, that the airport need not optimize the environment because the passenger will optimize for the airport. The high-stakes passenger does not.

The Zurich-Google question is the sharpest single objection in the Contrarian brief and deserves a direct answer. Zurich's wayfinding reputation predates Google Live View — its signage discipline and tenant-enforcement structure were documented in the European airport literature in the early 2000s, more than a decade before Google's indoor camera navigation reached the airport. Helsinki, Munich, Changi, and Frankfurt wayfind well without a comparable Google deployment. The pre-digital record stands [Aviation Historian Brief; Architectural Historian Brief]. The more interesting answer: the Contrarian is partially right about the present-day overlay, and the thesis is strengthened rather than weakened by saying so. Google Live View at Zurich works because it is layered onto a coherent physical signage hierarchy that does not contradict it. A digital overlay onto a contradictory physical system produces dueling sources of authority — exactly what passengers at US hubs experience when their phone tells them one thing and the overhead sign tells them another. Digital overlays inherit the governance quality of the physical system beneath them.

The capital-base counter-argument deserves its concession. Helsinki's €1 billion, Zurich's CHF 2.4 billion, Changi's S\$3 billion — these airports built unified physical environments. Their wayfinding clarity is in part the result. The Architectural Historian brief is direct: spatial legibility cannot be retrofitted with signs. A spatially incoherent terminal will defeat any sign system. The post-9/11 sterile-area boundary is an architectural condition, not merely a security procedure; it divided every terminal into two spatially distinct environments with no continuous spatial logic connecting them. Passengers spend most of their time in the zone the architect spent least time designing [Architectural Historian Brief]. This is the central wayfinding crisis of the contemporary airport, and it is older than any standards manual.

A signage program at IAD cannot make Eero Saarinen's Main Terminal into Norman Foster's Stansted. The Mobile Lounge / AeroTrain mismatch is not a sign problem at root; it is a circulation-architecture problem — a hub-and-spoke operational model retrofitted onto a building designed for a different circulation logic. United's preferred answer, extending the AeroTrain to Concourse D, would solve it. Signage cannot.

What signage can do is order the decision chain inside the existing architecture so each link works rather than fails. Better signs do not make IAD into Stansted; they make IAD's existing geometry as legible as it can be. The alternative is the geometry plus the contradiction debt, which is strictly worse. PANYNJ is the present-day evidence: a multi-airport retrofit, underway, visibly further along than any peer authority that has not undertaken the work. Five years from manual publication to prototype rollout across fourteen terminals is, in industry timeframes, fast. The counter-case requires the reader to believe that no governance solution is possible without a billion-dollar capital event preceding it. PANYNJ's actual progress does not support that claim.

The contractual cage is real. Majority-in-interest clauses are real. Renegotiating a use agreement against airline objection is a multi-year political act in which the carriers hold substantial power. None of this is in dispute. But the lease-structure argument concedes the point it most needs to defeat. The airport operator has unambiguous authority over common areas — ticketing halls, security queue approaches, baggage claim, ground transportation curbs, escalators, transit stations, the transition zones between airline-controlled concourses. These are the most critical decision points in the passenger journey. They are where the eye-tracking data says wayfinding failures concentrate. They are where the airport's authority is contractually unquestioned. The MII clause does not apply to operating standards classified appropriately; the FAA's distinction between "airport-installed" and "airline-installed" systems is a constraint on the airline-controlled zones, not on the airport's authority in the rest of the building [Airline Commercial Strategist Brief; Regulatory and Political Analyst Brief].

Will the dominant carrier support a wayfinding standard that covers its concourse? is the right question for the airline-controlled zones and the wrong question for the rest of the building. United will resist any standard that constrains its Polaris lounge environment, its gate holdroom branding, or its proprietary gate display technology. American will resist any blanket standard that asks it to subordinate the \$1 billion in capital it has just installed at DCA. Those resistances are real and substantial. They do not extend to the AeroTrain station, the National Hall, the baggage claim floor, the escalator at the bottom of which the connecting passenger is making her decision. Those are MWAA's.

The PANYNJ implementation timeline is not evidence that governance fails. It is evidence that governance takes time and political capital. The alternative — every terminal renovation generating a new signage layer, every carrier installation contradicting the last, every TSA reconfiguration carving out another exempt zone — is not a faster path. It is the path MWAA has been on for thirty years.

The Contrarian's depreciation point is correct and friendly to the thesis. Capital will be spent on signage regardless. The question is whether it is spent reactively, accumulating as patchwork, or spent strategically against a published standard. Every dollar of reactive signage spend across the next fifteen-year capital cycle is a dollar that could have purchased a system instead of a patch.

The staff-intervention point is also correct, and the answer is *both, in parallel*. Better PA tuning, gate agent positioning, information desk placement, FIDS refresh-rate calibration are valuable, fast, and below any reasonable MII threshold. They should be funded. They are not substitutes for signage rationalization. A staffed information desk is the system's exception-handling layer; better placement does not address the inconsistency that produces the exceptions. And the cognitive load eye-tracking documents at decision points is a property of the physical environment, not of the staff layer. A passenger at the bottom of an escalator with twenty seconds to choose a path is not helped by an agent positioned thirty yards away.

Dynamic digital systems do real work. FIDS is settled technology. Interactive kiosks at high-confusion points add value. Real-time security wait times reduce checkpoint anxiety. Multilingual digital assistance at international transfer zones reaches passengers that English-language static signs cannot. The thesis denies that *more digital* is, by itself, the answer. The widely cited \$700 million annual concession-revenue-loss figure is not independently auditable; the original methodology is not traceable; both the Infrastructure Economist and Technology Scout briefs flag it as vendor-sourced and unreliable. It should be set aside. The structurally valid point underneath it — that disoriented passengers under-spend and over-complain — has independent support from the 89-airport peer-reviewed elasticity study, properly discounted for wayfinding's share of total dwell time. That is the number the case should rest on. The case for digital depends on the same governance arrangement the thesis is arguing for. A digital wayfinding network with no enforcement of content hierarchy, no constraint on advertising co-location, no single owner of the standard, and no funded refresh cycle becomes the contradiction debt at higher capital intensity. Manchester Airport's documented improvement under a single Head of Visual Environment came in part from explicitly choosing not to use signage real estate for advertising in navigation-critical locations [Technology Scout Brief]. The lifecycle data argues for funded refresh discipline as a precondition for digital investment, not for less digital.

The Contrarian's sharpest procedural point is that the institutional reflex at most US airports, given a finding that wayfinding is a governance problem, will be to form a committee, produce a design standards document, and use the production of the document as evidence of action. The document will sit in a drawer. The airlines will continue to hang their own signs. TSA will continue to staple up its own notices. The capital programs will continue to layer new signage onto the old. The contradiction debt will continue to compound.

This risk is real, and any honest version of the thesis has to name it. The answer is that the deliverable is not a design document. The deliverable is an enforcement protocol. Tampa's Journey Segment Evaluation Tool is the model: a mechanism requiring every new capital project to be evaluated against the existing sign hierarchy at each decision point it touches, with a designated reviewer who has actual veto authority. PANYNJ's Aviation Customer Experience structure is the same idea at scale. Denver International's SignAgent deployment tracks 10,000 signs as a queryable database, reducing work-order generation from 4–5 days to minutes [Operations Analyst Brief; Chief Engineer Brief]. The committee risk is mitigated for an airport that treats wayfinding as an asset class with a named owner, a published lifecycle plan, and an enforcement protocol tied to every capital approval gate.

The defensible version of the asymmetry claim follows the brief math, not the slogan. The Infrastructure Economist brief calibrates a \$5–10M wayfinding program against \$6–9M in annual incremental concession revenue at MWAA's elasticity base — annual returns that, undiscounted across a ten-year horizon, imply a gross multiple in the high single digits relative to program cost, with additional return from reduced customer-service load, deferred sign-maintenance cost, and reduced service-recovery exposure.

That is excellent in capital terms. It is not the asymmetry by itself. The asymmetry lives on the cost side: the cost of wayfinding governance — roughly one to two percent of a single year's MWAA capital spend, depending on scope — is small relative to what wayfinding governs, which is the legibility of every square foot the rest of the capital program builds. A poorly wayfound terminal makes a billion-dollar concourse renovation harder to navigate; a well-wayfound one makes a fifty-year-old terminal usable. The cost is set against the value of every square foot the airport already owns and every square foot it will build for the next fifteen years. That is the asymmetry.

The strict version of the thesis: wayfinding is among the highest-yield, lowest-incremental-cost capital interventions available *in the airport's common-use areas*, with the lowest external-stakeholder dependency for the work that produces the most of the value, and with returns whose elasticities are documented in peer-reviewed literature. It is not the only cheap lever. It is the cheapest lever the airport can pull, in its common-use areas, without negotiating with someone who can say no.

VI. Implications for MWAA

A council writing for MWAA owes specific recommendations. What MWAA has, today, is two airports under a single authority operating across the most stratified signage history in the US large-hub inventory. IAD carries six construction eras and two ground-transport modes serving the same connection use case. DCA is fresh off Project Journey's \$1 billion capital program, including the five-day, 1,000-sign renaming, and is operating new American Airlines concourses built to American's specifications. The 2025–2039 Airport Use and Lease Agreement is a hybrid residual structure with a 1.40x debt service coverage target and a 1.25x bond covenant floor. Moody's projects MWAA's adjusted debt per O&D enplanement will peak at approximately \$400 in 2028, with net revenue DSCR declining from 1.63x in 2024 to roughly 1.3x over the next five years [Infrastructure Economist Brief]. United carries approximately two-thirds of IAD enplanements including United Express; American controls 53 percent of DCA flights. The \$9 billion capital program is set. The \$514,000 IAD Masterplan budget and the \$4.87 million DCA equivalent are study budgets, not implementation programs [Chief Engineer Brief].

The ADA 2027 deadline is the forcing function

The single most time-bounded fact in this report is that ADA Title II's compliance date for digital accessibility was extended to April 26, 2027 for larger entities — approximately ten months from the date of this draft [Chief Engineer Brief]. MWAA's existing FIDS, BIDS, dynamic directional, and interactive kiosk inventory will require accessibility compliance by that date. The deadline turns "wayfinding is overdue" into "wayfinding is timed against an external compliance forcing event." Any digital-component refresh undertaken in 2026 to comply with the 2027 deadline that is not coordinated with a standards manual will produce a new contradiction layer the standards manual will then have to retire. The same money spent inside a forward standard produces a system. The compliance deadline is the lever. It will not return.

The operational consequence is that the standards manual schedule has to be inverted. The masterplan studies already budgeted do not need to finish before the digital-accessibility work begins. They need to finish *first* on the rules that govern the accessibility work — content hierarchy, character height, contrast ratios, audio-loop coverage, screen-reader compatibility for interactive kiosks. Those are the same rules ACRP 52, AC 150/5360-12F, and the 2027 ADA digital accessibility standard variously require, and they are auditable in months, not years. The full standards manual can complete on its own schedule. The accessibility-relevant rules need to be promulgated as Phase 1 by Q4 2026 to govern the 2027 spend.

A named owner, on the PANYNJ model

Treat wayfinding as an asset class with a single owner reporting at the executive level. Not a committee. Not a working group. A program manager for wayfinding, reporting to a senior MWAA executive at the central office on the PANYNJ Aviation Customer Experience model, with explicit veto authority over signage in MWAA-controlled common areas at both airports — and a published Wayfinding and Signing Standards Manual covering IAD and DCA under a single set of rules. MWAA's Design Manual currently runs to seven volumes, with separate tenant design standards for IAD's Main Terminal / Concourse Z and Concourse B — two parts of the same airport published as two distinct documents [Regulatory and Political Analyst Brief]. The volume count is evidence of the problem.

The naming matters because the absence of a name is what produces the committee. PANYNJ named the position. MWAA can name it. The role is not organizational expansion; it is the assignment of authority currently dispersed across several existing positions in Engineering, Operations, and the Vice President's office, none of which has it as a primary responsibility and none of which can override the others.

Move from study to implementation, and structure to live under MII

The \$514,000 IAD allocation and the \$4.87 million DCA allocation are appropriate for the study phase. The implementation phase is missing. A defensible scope estimate, calibrated from international consultancy-framework comparables adjusted for US construction conditions, runs \$15 to \$25 million for a phased rationalization across both airports. The honest accounting: Heathrow's 2013 £10 million five-year framework was a consultancy and ad hoc signage project budget, not a hardware-and-installation program. Inflated to 2026 dollars at roughly 50 percent UK construction inflation, the consultancy base is approximately \$19 million; scaled by MWAA's 52 million passengers against Heathrow's 80 million produces a baseline near \$12 million. Two adjustments the Heathrow figure does not include: the Chief Engineer's 40–50 percent live-terminal construction premium adds \$5 to \$6 million on the installation share, and a contingency reserve calibrated against the Ontario International 2.7x overrun multiplier — applied at a discounted rate across a multi-project program rather than a single project — implies a \$4 to \$5 million reserve. The total is an order-of-magnitude range, not a derived budget. Precise scoping is downstream of the masterplan studies already budgeted [Infrastructure Economist Brief; Chief Engineer Brief; analyst calculation].

A \$15 to \$25 million program will cross the typical \$5–\$10 million MII threshold by any reasonable accounting. The strategy is not to claim the program automatically sidesteps MII; it is to classify aggressively. Capital classification should be reserved for genuinely infrastructure-grade elements — sign substrates, structural attachments, digital backbone, asset management platform. The dominant share of program cost — design, standards governance, sign fabrication, installation, change management — is appropriately operating expense and should be funded from the airport's operating budget rather than the capital plan. The structuring is aggressive but defensible. Airlines may challenge it. The structuring strategy is the work; the standard is downstream of the structuring.

The PFC cap, unchanged since 2001 and now worth roughly half its 2001 purchasing power, is the underlying reason discretionary capital will not appear for a wayfinding program of this size [Regulatory and Political Analyst Brief]. Funding from operating budget against a multi-year operating program is not a workaround. It is the only path available. Treating wayfinding as an operating discipline is partly a governance choice; it is also a funding-mechanics necessity.

The IAD transport-mode decision chain

Solve the IAD transport-mode wayfinding problem at the decision point, with the honest concession that signage alone cannot fully resolve it. The Airline Commercial Strategist brief is precise: the AeroTrain–Mobile Lounge transition is a decision chain, not a single decision point, and each link is a wayfinding problem [Airline Commercial Strategist Brief]. The signage solution orders the chain so each link works rather than fails; it does not eliminate the underlying mode mismatch. The intervention is to position the FIDS-first wayfinding decision at the transit-mode interface, route the passenger by mode rather than by concourse, and provide consistent directional confirmation along the Mobile Lounge bay. The first-best fix is extending the AeroTrain to Concourse D — United's preferred solution, a multi-hundred-million-dollar capital program, currently unfunded. The signage intervention is second-best, an order of magnitude cheaper, while the first-best is unfunded and not on a credible near-term schedule. Second-best at order-of-magnitude cheaper is the right call.

The Saarinen catenary-cable roof is a constraint to design around rather than argue against. The IAD signage standard should specify that overhead pendant signage in the Main Terminal is a constrained inventory — every existing pendant is presumed to be in its final position, and additions require a formal exception process. Floor-mounted directional cues, wall-mounted directories, and digital displays at landside-airside transition points should carry the navigation load that would, in a less constrained building, be carried by overhead pendants. The Architectural Historian brief is direct: the Saarinen processional logic was broken when the AeroTrain replaced the Mobile Lounge as the dominant transport mode; the IAD wayfinding problem is the residue of that change. The forward standard cannot restore the processional logic. It can prevent the next capital event from depositing another contradictory layer on top of it [Architectural Historian Brief].

Validation and the asset platform

Budget for eye-tracking validation before and after implementation, and fund an asset-management platform on the SignAgent model. A terminal-scale eye-tracking study using the Tobii methodology runs \$40,000 to \$80,000 [Technology Scout Brief] — a rounding error against the implementation budget. Any wayfinding program installed without pre- and post-validation is a program that does not know whether it worked. Denver International's SignAgent deployment — 10,000 signs as a queryable asset database, work-order generation reduced from 4–5 days to minutes — turns a standards manual from a document into a system [Chief Engineer Brief]. MWAA's existing Pointr indoor positioning contract (six-figure, 2019) is data without a frame [Operations Analyst Brief]; an asset management platform is the frame.

What this is not. It is not a typeface recommendation. The graphic designers will produce competent design. The competent design is not the problem MWAA has. The problem is that the design, once produced, has historically been undermined by every adjacent stakeholder and every subsequent capital project. The forward standard MWAA needs is the one the next capital project, the next federal mandate, and the next carrier renegotiation cannot quietly displace. That standard rests on three things: an owner with named authority, an asset management discipline that makes the system auditable, and an enforcement protocol embedded in the gate review process for every project that touches a sign. The pieces are commercially available. The decision is institutional.

The connecting passenger from the opening of this report will, if MWAA does the work, step off her aircraft and find the answer she needs at the first decision point she encounters. She will not know whose name is on the standards document. She will not know that the sign she is reading is a different generation from the one above her gate. She will read it, choose the right mode, and make her connection.

If MWAA does not do the work, the next capital project will deposit one more layer, the next federal mandate will create one more exempt zone, and the building she walked into will become marginally harder to read than the one her predecessor walked into ten years ago.

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